

STT 3850 : Week 7

Spring 2025

Appalachian State University

Section 1

Outline for the week

By the end of the week: Foundations of Probability

- Laws of probability
- Conditional probability
- Law of Total Probability
- Bayes' Rule

Section 2

Laws of probability

Basics of probability

- The set of all possible outcomes of a random experiment is called the sample space, Ω . An event E is a subset of Ω .
- Note: event E can be an outcome or a combination of outcomes.
- The probability of an event, E , is its **long term relative frequency**.

Basics of probability

- If the probability is based on repeatedly observing the event's outcome, this probability is called **empirical probability** and given by:

$$\mathbb{P}(E) = \frac{\text{Number of times } E \text{ occurs}}{\text{Total number of trials}}$$

- When the probability comes from a mathematical model and not from observations, it is called **theoretical probability**. This is given by:

$$\mathbb{P}(E) = \frac{\text{Number of outcomes in } E}{\text{Number of possible outcomes}}$$

Axioms of probability

- ① For any event E : $0 \leq \mathbb{P}(E) \leq 1$
- ② $\mathbb{P}(\Omega) = 1$
- ③ $\mathbb{P}(E) + \mathbb{P}(E^c) = 1$ so $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$

We define E^c as a complement of E . E^c is the event of E not happening.

- ④ If E and F can not happen at the same time (mutually exclusive or disjoint), then

$$\mathbb{P}(E \cup F) = \mathbb{P}(E \text{ or } F) = \mathbb{P}(E) + \mathbb{P}(F)$$

In general, for any sequence of mutually exclusive events $E_1, E_2, \dots$ (that is $E_i \cap E_j = \emptyset$ for all $(i \neq j)$)

$$\mathbb{P}(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$$

Example 1

In a dresser, there are 7 blue shirts, 5 red shirts, and 8 black shirts.

- (a) What is the probability of randomly selecting a red shirt?
- (b) What is the probability that a randomly selected shirt is not black?

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Solution.

Example 2: Birthday Problem

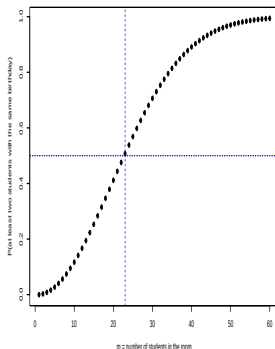
Suppose that a room contains m students. What is the probability that at least two of them have the same birthday? Start by assuming every day of the year is equally likely to be a birthday, and disregard leap years. That is, assume there are always $n = 365$ days to a year.

Example 2: Birthday Problem

Solution:

Some R

```
m <- 1:60                      # vector of number of students
p <- numeric(60)               # initialize vector to 0's
for(i in m){                   # index value for loop
  q = prod((365:(365 - i + 1))/365)
  p[i] = 1 - q
}
plot(m, p, pch = 19, ylab = "P(at least two students with the same birthday)",
     xlab = "m = number of students in the room")
abline(h = 0.5, lty = 2, col = "blue")
abline(v = 23, lty = 2, col = "blue")
```



Addition Rules for Probability

- If E and F can not happen at the same time (mutually exclusive or disjoint), then

$$\mathbb{P}(E \cup F) = \mathbb{P}(E \text{ or } F) = \mathbb{P}(E) + \mathbb{P}(F)$$

In general, for any sequence of mutually exclusive events $E_1, E_2, \dots$ (that is $E_i \cap E_j = \emptyset$ for all $(i \neq j)$)

$$\mathbb{P}(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$$

- For any two events E and F ,

$$\mathbb{P}(E \cup F) = \mathbb{P}(E \text{ or } F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F)$$

Example 3

What is the probability of drawing a diamond or a spade from a standard card deck?

Solution:

Example 4

What is the probability of drawing a diamond or an ace from a standard card deck?

Solution:

Example 5

A class with 30 students consists of 16 Math majors, 21 CS majors and 8 double CS and Math majors. If a student is randomly selected, what is the probability that the student is a Math major or a CS major?

Solution:

Multiplication Rules

- If E and F are independent

$$\mathbb{P}(E \cap F) = \mathbb{P}(E \text{ and } F) = \mathbb{P}(E) \times \mathbb{P}(F)$$

Example 6

What is the probability of selecting 2 aces if 2 cards are randomly selected with replacement?

Solution:

Conditional Probability and the General Multiplication Rule

- Another probability we want to be able to find is of the form “given F , what is the probability of E ?” This is known as a **conditional probability** and is written

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}$$

- Rearranging the conditional probability equation, we get the **General Multiplication Rule**:

$$\mathbb{P}(E \cap F) = \mathbb{P}(F) \times \mathbb{P}(E|F)$$

Equivalently,

$$\mathbb{P}(E \cap F) = \mathbb{P}(E) \times \mathbb{P}(F|E)$$

Definition of Independence

- Events E and F are independent if knowing E happened does not change the probability of F .

$$\mathbb{P}(E \cap F) = \mathbb{P}(E) \times \mathbb{P}(F)$$

Then:

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} = \frac{\mathbb{P}(E) \times \mathbb{P}(F)}{\mathbb{P}(F)} = \mathbb{P}(E)$$

- If E and F are independent then $\mathbb{P}(E|F) = \mathbb{P}(E)$
- Equivalently

$$\mathbb{P}(F|E) = \mathbb{P}(F)$$

Example 7

What is the probability of selecting 2 aces if 2 cards are randomly selected without replacement?

Solution:

Example 8

Suppose two fair dice are rolled where each of the 36 possible outcomes is equally likely to occur. Knowing that the first die shows a 3, what is the probability that the sum of the two dice equals 6?

Solution:

Section 3

Law of Total Probability and Bayes' Rule

Law of Total Probability and Bayes' Rule

- Bayes' Rule is used for reversing the condition
 - The formula is easier with the tree!
- In general: Let $F_1, F_2, \dots, F_n$ be such that $\cup_{i=1}^{\infty} F_i = \Omega$ and $F_i \cap F_j = \emptyset$ for all $i \neq j$, with $\mathbb{P}(F_i) > 0$ for all i . Then, for any event E

$$\mathbb{P}(E) = \sum_{i=1}^n \mathbb{P}(E \cap F_i) = \sum_{i=1}^n \mathbb{P}(E|F_i)\mathbb{P}(F_i)$$

$$\mathbb{P}(F_i|E) = \frac{\mathbb{P}(E \cap F_i)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E|F_i)\mathbb{P}(F_i)}{\sum_{i=1}^n \mathbb{P}(E|F_i)\mathbb{P}(F_i)}.$$

Example 1

Suppose the prior probabilities that a document was written in Word, LaTeX, and HTML are: 0.50, 0.30, and 0.20, respectively. From past experience it is known that 40% of Word documents exceed 10 pages; 20% of LaTeX documents exceed 10 pages; 20% of HTML documents exceed 10 pages.

- What is the overall proportion of documents containing more than 10 pages?
- If the document that was chosen at random was found to exceed 10 pages, what is the probability it was written in LaTeX?
- If the document that was chosen at random was found to exceed 10 pages, what is the probability it was written in Word?
- If the document that was chosen at random was found to exceed 10 pages, what is the probability it was written in HTML?

Example 1 - Partial Solution

- Let E be the event a document selected at random is more than 10 pages.
- Let W be the event the document is written with Word.
- Let L be the event the document is written with LaTeX.
- Let H be the event the document is written with HTML.

$$E = (W \cap E) + (L \cap E) + (H \cap E)$$

$$\begin{aligned}\mathbb{P}(E) &= \mathbb{P}(W \cap E) + \mathbb{P}(L \cap E) + \mathbb{P}(H \cap E) \\ &= \mathbb{P}(W)\mathbb{P}(E|W) + \mathbb{P}(L)\mathbb{P}(E|L) + \mathbb{P}(H)\mathbb{P}(E|H) \\ &= (0.5 \times 0.4) + (0.30 \times 0.20) + (0.2 \times 0.2) = 0.30\end{aligned}$$

The probability a randomly selected document is more than 10 pages is 0.30.

Example 1 - Partial Solution

If the document that was chosen at random was found to exceed 10 pages, what is the probability it was written in LaTeX?

$$\begin{aligned}\mathbb{P}(L|E) &= \frac{\mathbb{P}(L) \cdot \mathbb{P}(E|L)}{\mathbb{P}(E)} \\&= \frac{\mathbb{P}(L) \cdot \mathbb{P}(E|L)}{\mathbb{P}(W \cap E) + \mathbb{P}(L \cap E) + \mathbb{P}(H \cap E)} \\&= \frac{\mathbb{P}(L) \cdot \mathbb{P}(E|L)}{\mathbb{P}(W)\mathbb{P}(E|W) + \mathbb{P}(L)\mathbb{P}(E|L) + \mathbb{P}(H)\mathbb{P}(E|H)} \\&= \frac{0.3 \times 0.2}{0.5 \times 0.4 + 0.3 \times 0.2 + 0.2 \times 0.2} \\&= \frac{0.06}{0.3} = 0.2\end{aligned}$$

Given a randomly selected document was found to exceed 10 pages, the probability it was written in LaTeX is 0.2.

Example 1 - Partial Solution

If the document that was chosen at random was found to exceed 10 pages, what is the probability it was written in Word?

$$\begin{aligned}\mathbb{P}(W|E) &= \frac{\mathbb{P}(W) \cdot \mathbb{P}(E|W)}{\mathbb{P}(E)} \\&= \frac{\mathbb{P}(W) \cdot \mathbb{P}(E|W)}{\mathbb{P}(W \cap E) + \mathbb{P}(L \cap E) + \mathbb{P}(H \cap E)} \\&= \frac{\mathbb{P}(W) \cdot \mathbb{P}(E|W)}{\mathbb{P}(W)\mathbb{P}(E|W) + \mathbb{P}(L)\mathbb{P}(E|L) + \mathbb{P}(H)\mathbb{P}(E|H)} \\&= \frac{0.5 \times 0.4}{0.5 \times 0.4 + 0.3 \times 0.2 + 0.2 \times 0.2} \\&= \frac{0.2}{0.3} = \frac{2}{3}\end{aligned}$$

Given a randomly selected document was found to exceed 10 pages, the probability it was written in Word is $2/3$.

Example 1 - Partial Solution

If the document that was chosen at random was found to exceed 10 pages, what is the probability it was written in HTML?

$$\begin{aligned}\mathbb{P}(H|E) &= \frac{\mathbb{P}(H) \cdot \mathbb{P}(E|H)}{\mathbb{P}(E)} \\&= \frac{\mathbb{P}(H) \cdot \mathbb{P}(E|H)}{\mathbb{P}(W \cap E) + \mathbb{P}(L \cap E) + \mathbb{P}(H \cap E)} \\&= \frac{\mathbb{P}(H) \cdot \mathbb{P}(E|H)}{\mathbb{P}(W)\mathbb{P}(E|W) + \mathbb{P}(L)\mathbb{P}(E|L) + \mathbb{P}(H)\mathbb{P}(E|H)} \\&= \frac{0.2 \times 0.2}{0.5 \times 0.4 + 0.3 \times 0.2 + 0.2 \times 0.2} \\&= \frac{0.04}{0.3} = 0.1333333\end{aligned}$$

Given a randomly selected document was found to exceed 10 pages, the probability it was written in HTML is 0.1333333.

Example 2

Dan's Diner employs three dishwashers. Al washes 20% of the dishes and breaks only 1% of those he handles. Betty and Chuck each wash 40% of the dishes, and Betty breaks only 1% of hers, but Chuck breaks 5% of the dishes he washes. You go to Dan's for supper one night and hear a dish break at the sink. What's the probability that Chuck is on the job?

Example 2

Solution:

Choose a Door

The television show Let's Make a Deal hosted by Monty Hall, gave contestants the opportunity to choose one of three doors. Contestants hoped to choose the one that concealed the grand prize. Behind the other two doors were much less valuable prizes. After the contestant chose one of the doors, say Door 1, Monty opened one of the other two doors, say Door 3, containing a much less valuable prize. The contestant was then asked whether he or she wished to stay with the original choice (Door 1) or switch to the other closed door (Door 2). What should the contestant do? Is it better to stay with original choice or switch to the other closed door? (a) What is the probability of winning by switching doors when given the opportunity? (b) What is the probability of winning by staying with the initial selection?

Choose a Door: R Code

```
set.seed(13)
n <- 100000
actual <- sample(1:3, size = n, replace = TRUE)
aguess <- sample(1:3, size = n, replace = TRUE)
equals <- (actual == aguess)
not.eq <- (actual != aguess)
PnoSwitch <- mean(equals)
PwiSwitch <- mean(not.eq)
probs <- c(PnoSwitch, PwiSwitch)
names(probs) <- c("P(Win no Switch)", "P(Win Switch)")
probs
```

P(Win no Switch)	P(Win Switch)
0.3276	0.6724

Choose a Door: Solution

To solve with Bayes' Rule, start by assuming the contestant initially guesses Door 1 and that Monty opens Door 3. Let the event $D_i = \text{Door } i \text{ conceals the prize}$ and $O_j = \text{Monty opens door } j \text{ after the contestant selects Door 1}$. When a contestant initially selects a door, $\mathbb{P}(D_1) = \mathbb{P}(D_2) = \mathbb{P}(D_3) = 1/3$. (Similar reasoning works independent of the originally selected door.) Once Monty shows the grand prize is not behind Door 3, the probability of winning the grand prize is now one of $\mathbb{P}(D_1|O_3)$ or $\mathbb{P}(D_2|O_3)$. Note that $\mathbb{P}(D_1|O_3)$ corresponds to the strategy of sticking with the initial guess and $\mathbb{P}(D_2|O_3)$ corresponds to the strategy of switching doors. Based on how the show is designed, the following are known:

Choose a Door: Solution

- $\mathbb{P}(O_3|D_1) = 1/2$ since Monty can open one of either Door 3 or Door 2 without revealing the grand prize.
- $\mathbb{P}(O_3|D_2) = 1$ since the only door Monty can open without revealing the grand prize is Door 3.
- $\mathbb{P}(O_3|D_3) = 0$ since Monty will not open Door 3 if it contains the grand prize.

Choose a Door: Solution

$$\begin{aligned}\mathbb{P}(D_1|O_3) &= \frac{\mathbb{P}(O_3|D_1)\mathbb{P}(D_1)}{\mathbb{P}(O_3|D_1)\mathbb{P}(D_1) + \mathbb{P}(O_3|D_2)\mathbb{P}(D_2) + \mathbb{P}(O_3|D_3)\mathbb{P}(D_3)} \\ &= \frac{1/2 \times 1/3}{1/2 \times 1/3 + 1 \times 1/3 + 0 \times 1/3} = \frac{1}{3}.\end{aligned}$$

$$\begin{aligned}\mathbb{P}(D_2|O_3) &= \frac{\mathbb{P}(O_3|D_2)\mathbb{P}(D_2)}{\mathbb{P}(O_3|D_1)\mathbb{P}(D_1) + \mathbb{P}(O_3|D_2)\mathbb{P}(D_2) + \mathbb{P}(O_3|D_3)\mathbb{P}(D_3)} \\ &= \frac{1 \times 1/3}{1/2 \times 1/3 + 1 \times 1/3 + 0 \times 1/3} = \frac{2}{3}.\end{aligned}$$

Therefore, it is always to the contestant's benefit to switch doors.

Monte Hall Problem with n Doors

- Consider the case with n Doors and just one grand prize
 - The probability of winning the grand prize on the first choice is $\frac{1}{n}$.
 - This is the same probability of winning if you were to use the “stay” strategy.
- You will win the grand prize if both of the following events occur:
 - You choose a door that does not have the grand prize. Note:
 $\mathbb{P}(G) = \frac{n-1}{n}$.
 - You choose the door with the grand prize by switching doors
 $\mathbb{P}(C|G) = \frac{1}{n-2}$.

Monte Hall Problem with n Doors

In other words, the probability of winning using the “switch” strategy is

$$\mathbb{P}(G \cap C) = P(G) \cdot \mathbb{P}(C|G) = \frac{n-1}{n} \cdot \frac{1}{n-2} = \frac{n-1}{n(n-2)}$$

For $n = 3$ doors, the probability of winning with the “switch” strategy is:

$$\frac{n-1}{n(n-2)} = \frac{3-1}{3(3-2)} = \frac{2}{3}$$

Note:

$$\frac{n-1}{n(n-2)} > \frac{1}{n}$$

This implies switching is always better.